# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## CLASS 10 — SCIENCE CHAPTER TEST SET B

Chapter 10: **The Human Eye and the Colourful World** | RBSE Board Pattern | Session 2026–27

**Time: 1 Hour 30 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 40**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

## SECTION — A (Objective Type Questions) 10 × 1 = 10

*Choose the correct option and write it in your answer book.*

**1.** The image formed on the retina of the eye is: 1

**(a)** virtual and inverted **(b)** real and erect
**(c)** virtual and erect **(d)** real and inverted

**2.** The change in focal length of the eye lens is caused by: 1

**(a)** ciliary muscles **(b)** pupil **(c)** retina **(d)** cornea

**3.** Hypermetropia is corrected using a: 1

**(a)** plane glass **(b)** concave lens **(c)** prism **(d)** convex lens

**4.** A student states: “Violet light has the longest wavelength.” The correct response is: 1

**(a)** Correct
**(b)** Correct in water
**(c)** Incorrect — red has the longest wavelength
**(d)** Cannot be determined

**5.** For a rainbow, Student A says it is seen opposite the Sun; Student B says it is seen towards the Sun. Then: 1

**(a)** Student B is correct **(b)** Student A is correct
**(c)** Both are correct **(d)** Both are wrong

**6.** The opening through which light enters the eye is the: 1

**(a)** lens **(b)** pupil **(c)** retina **(d)** iris

**7.** The light-sensitive screen of the eye is the: 1

**(a)** iris **(b)** retina **(c)** cornea **(d)** lens

**8.** The phenomenon responsible for the blue colour of the sky is: 1

**(a)** reflection **(b)** dispersion
**(c)** scattering of light **(d)** refraction

**9.** The defect corrected by bifocal lenses is: 1

**(a)** hypermetropia only **(b)** cataract
**(c)** myopia only **(d)** presbyopia

**10.** The sequence of colours in a spectrum from the base of the prism is: 1

**(a)** red to violet **(b)** green to red **(c)** violet to red **(d)** blue to orange

## SECTION — B (Very Short Answer Type) — 4 × 2 = 8

*Answer in about 10–15 words each.*

**11.** What is the function of the iris? 2

**12.** Why does the pupil become small in bright light? 2

**13.** What is the Tyndall effect? 2

**14.** A student says that a person with myopia uses a convex lens. Justify or refute. 2

## SECTION — C (Short Answer Type) — 4 × 3 = 12

*Answer in about 30–40 words each.*

**15.** What is the power of accommodation of the eye? Why does it decrease with age? 3

**16.** Why is a normal eye not able to see clearly objects placed closer than 25 cm? 3

**17.** Draw (describe) a ray diagram to show how a concave lens corrects myopia. 3

OR

Why does the sky appear dark to astronauts? Why does the sky appear blue to us? 3

**18.** Explain the formation of a rainbow with a diagram (describe). 3

OR

Why do stars appear higher than their actual position? 3

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**19.** (i) What is hypermetropia? 5
(ii) Describe with ray diagrams the defect and its correction.
(iii) What are the causes?

OR

(i) Explain atmospheric refraction with two examples. 5
(ii) Why does the apparent position of a star change?
(iii) Why do the objects seen through hot air appear to waver?

**20.** **Case study:** An old man reading a newspaper holds it at arm's length and also cannot see distant boards clearly. 5
(i) Name the defect.
(ii) What causes it?
(iii) What type of lens is prescribed?
(iv) How is it designed?
(v) Name a surgical option for eye defects.

OR

(i) What is a spectrum? 5
(ii) Why does white light split in a prism?
(iii) Draw (describe) the arrangement to recombine the spectrum.
(iv) Name the colour with maximum deviation.

## END OF QUESTION PAPER — Total: 40 Marks